

Supplementary *on line* material for the paper "Propensity score matching with clustered data. An application to the estimation of the impact of caesarean section on the Apgar score"

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In this document we provide additional material for the paper *Propensity score matching with clustered data. An application to the estimation of the impact of caesarean section on the Apgar score*. Section A provides the exact formulas for equations (7) and (8) used to generate the true treatment and outcome data in the simulation studies. Section B provides the R code used in the baseline simulations. Section C includes full tables of simulation results not included in the paper for the sake of saving space. Copyright © 2010 John Wiley & Sons, Ltd.

A - Equations for the data generating models

In this Section we report the exact formulas of equations (7) and (8) in section 4 of the main text. As explained in the text, the coefficients of individual-level covariates in the true models generating the treatment and the outcome were set to values similar to observed coefficients estimated on the real data.

In particular, the probability of being treated (7) was generated according to:

$$\begin{aligned} e_{ij} &= 1/[1 + \exp(\beta_0 + \beta_1 X_{1ij} + \cdots + \beta_k X_{kij} + \beta_Z Z_j) + u_j] \\ &= 1/[1 + \exp(-1.69 + 0.46 \cdot X_1 + 0.54 \cdot X_2 + 0.74 \cdot X_3 + \\ &\quad + 1.15 \cdot X_4 + 0.08 \cdot X_5 - 0.03 \cdot X_6 + 0.33 \cdot X_7 + \\ &\quad + 0.85 \cdot X_8 + 0.04 \cdot X_9 - 0.2 \cdot X_{10} + \\ &\quad + 0.65 \cdot X_{11} + \beta_Z Z) + u_j] \end{aligned}$$

where β_Z varied in the set $\{0, 0.2, 0.4, 0.6\}$. The outcome (8) was generated according to:

$$\begin{aligned} P(Y_{ij} = 1) &= 1/[1 + \exp(\gamma_0 + \gamma_1 X_{1ij} + \cdots + \gamma_k X_{kij} + \gamma_Z Z_j + \alpha T_{ij} + v_j)] \\ &= 1/[1 + \exp(-5.35 + 0.02 \cdot X_1 - 0.12 \cdot X_2 - 0.20 \cdot X_3 + \\ &\quad - 0.47 \cdot X_4 + 0.11 \cdot X_5 + 0.12 \cdot X_6 + 0.58 \cdot X_7 + \\ &\quad - 1 \cdot X_8 - 0.38 \cdot X_9 - 0.2 \cdot X_{10} + \\ &\quad + 0.15 \cdot X_{11} + 0.4 \cdot Z + 0.41 \cdot T) + v_j]. \end{aligned}$$

The error terms at the cluster-level, u_j and v_j are generated as realizations of independent normal variables with zero mean and standard deviation equal to 0.2. As for the distribution of individual-level covariates we used the empirical distributions observed in the data for the variables we described in section 5 of the main text. More specifically:

$X_1 \sim$ maternal age (15-24 years)

$X_2 \sim$ maternal age (25-29 years)
 $X_3 \sim$ maternal age (30-35 years)
 $X_4 \sim$ maternal age (> 35 years)
 $X_5 \sim$ maternal education (high school)
 $X_6 \sim$ maternal education (graduate or more)
 $X_7 \sim$ infant weight (< 2500 grams)
 $X_8 \sim$ infant weight (> 4000 grams)
 $X_9 \sim$ induction of labour (yes=1, no=0)
 $X_{10} \sim$ gestational age (in weeks)
 $X_{11} \sim$ pregnancy related pathologies (yes=1, no=0).

Finally, the cluster-level variable was generated as $Z \sim N(0, 0.25)$.

B - Simulation code

In this Section we report the R code used for the baseline simulations. As far as was possible, we used the same notation (variables names, parameters name, etc.) as in the main text.

```
##### Read empirical data containing covariates
source("read data.R")

##### Simulation code

# Generate the cluster-level variable Z

d$Z<-rep(rnorm(nlevels(d$Hospital),0,0.25),table(d$Hospital))

# Set replacement

replacement<-TRUE

# Set caliper

caliper=0.20

# Set betaZ: coefficient of Z in treatment equation

for (betaZ in c(0,0.2,0.4,0.6)){

# Set number of iterations

R<-500

# Create storage vectors

sim.asams.noZ<-sim.asams.Z<-sim.asams
<-sim.drop<-pdrop<-sim.att<-sim.relatt
<-sim.se<-sim.ar<-sim.icc<-list()

# Start cycle

for (r in 1:R){

pb <- txtProgressBar(min = 0, max = R, style = 3)
setTxtProgressBar(pb, r)

# Generate random effects at the hospital level for the treatment equation
set.seed(r+6)
d$ret<-rep(rnorm(nlevels(d$Hospital),0,0.2),table(d$Hospital))

# Generate random effects at the hospital level for the outcome equation
set.seed(r+9)
d$rey<-rep(rnorm(nlevels(d$Hospital),0,0.2),table(d$Hospital))

# TREATMENT MODEL:
d$sim.probs<-
```

```
(1 + exp( -(4.96 + 0.46*d$X1 + 0.54*d$X2
+ 0.74*d$X3 + 1.15*d$X4 + 0.08*d$X5
- 0.03*d$X6 + 0.33*d$X7 + 0.85*d$X8
+ 0.04*d$X9 - 0.18*d$X10 + 0.65*d$X11
+ betaZ*d$Z + d$ret
) ) ) ^ -1

# Generate simulated treatment
set.seed(r)
unif <- runif(dim(d)[1], 0, 1)
d$sim.cs <- ifelse(d$sim.probcs > unif, 1, 0)

# OUTCOME MODEL:
d$sim.problowapgt1 <-
(1 + exp( -(8.46 + 0.02*d$X1 - 0.12*d$X2
- 0.20*d$X3 - 0.47*d$X4 + 0.11*d$X5
+ 0.12*d$X6 + 0.58*d$X7 - 1*d$X8
- 0.38*d$X9 - 0.34*d$X10 + 0.15*d$X11
+ 0.41sim.cs + 0.4*d$Z + d$rey
) ) ) ^ -1

# Generate simulated outcome
set.seed(r+1)
unif2 <- runif(dim(d)[1], 0, 1)
d$sim.lowapg <- ifelse(
(d$sim.problowapgt1 > unif2), 1, 0)

##### Propensity scores with logit model (strategies: NV, W, PW,)

d$sim.ps <- d$sim.logps
# logit ps
sim.fit1 <- glm(formula = sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11,
data = d, family = binomial(link = "logit"))
d$sim.ps <- fitted(sim.fit1)

##### Crude estimate:

trueatt <- mean(d$sim.problowapgt1[d$sim.cs == 1])
- mean(d$sim.problowapgt0[d$sim.cs == 1])

# ATT and SE

attbef <- mean(d$sim.lowapg[d$sim.cs == 1])
- mean(d$sim.lowapg[d$sim.cs == 0])

relattbef <- abs(attbef - trueatt) / trueatt

p0 <- (length(d$sim.lowapg[d$sim.cs == 1]) * mean(d$sim.lowapg[d$sim.cs == 1])
+ length(d$sim.lowapg[d$sim.cs == 0]) * mean(d$sim.lowapg[d$sim.cs == 0]))
/ dim(d)[1]

sebef <- sqrt( (p0 * (1 - p0) * (1 / length(d$sim.lowapg[d$sim.cs == 1]))
```

```
+1/length(d$sim.lowapg[d$sim.cs==0]))))

# Balance crude:

bal = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 + Z,
data=d, ks = FALSE, nboots=0, print.level=0)

asam_bef <-vector()
for(i in 1:18){asam_bef[[i]] <- bal$BeforeMatching[[i]]$sdiff}

##### Naive matching (NV)

rr = Match(
Y=d$sim.lowapg, Tr=d$sim.cs,
X=d$sim.ps, caliper=caliper,
M=1, replace=replacement,ties=TRUE)

#drops
mndrop<-table(d[rr$index.dropped,]$ord.Hospital)

# ATT and SE naive matching

attNV<-rr$est

relattNV<-abs(attNV-trueatt)/trueatt

m0<-lm(formula = sim.lowapg ~ sim.cs,
data=md, weights=c(rr$weights,rr$weights))#summary(m0)
m0.vcovCL<-cluster.vcov(m0, md$Hospital)
seNV<-coefest(m0, m0.vcovCL)[4]

# Average number of replicates of control units in matched dataset

ff<-(aggregate(rr$weights ~ rr$index.control,
data = cbind(rr$weights,rr$index.control), sum))

arA<-sum(ff[, "rr$weights"][which(ff[, "rr$weights"]>1)]-1)/
length(unique(ff[, "rr$index.control"]))#in general

# Balance after naive matching (NV):

bal_apm = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 + Z
, data=d, match.out=rr, ks = FALSE, nboots=0, print.level=0)

asamNV <-vector()
for(i in 1:18){asamNV[[i]] <- bal_apm$AfterMatching[[i]]$sdiff}

##### Within-cluster matching (W)

cumfreq<-c(0,cumsum(table(d$ord.Hospital)))
```

```
inddrop<-vector(); indtreat<-vector(); indcontr<-vector()
weights<-vector(); indatt<-vector() ; indse<-vector()

#the following cycle matches treated within hospitals

for( i in 1:(nlevels(d$Hospital) - 2))
{
  rr <- Match(
Y      =d[d$Hospital==levels(d$Hospital)[ i ],]$sim.lowapg,
Tr      =d[d$Hospital==levels(d$Hospital)[ i ],]$sim.cs,
X      =d[d$Hospital==levels(d$Hospital)[ i ],]$sim.ps,
caliper=caliper*sd(d$sim.ps)
/sd(d[d$Hospital==levels(d$Hospital)[ i ],]$sim.ps),
M=1, replace=replacement,ties=TRUE)

inddrop <-c(inddrop,cumfreq[i]+rr$index.dropped)
indtreat<-c(indtreat,cumfreq[i]+rr$index.treated)
indcontr<-c(indcontr,cumfreq[i]+rr$index.control)
indatt<-c(indatt,rr$est)
indse<-c(indse,rr$se)
weights<-c(weights,rr$weights)
}
mwndrop<-table(d[inddrop,]$Hospital) # all drops

mwd<-rbind(d[indtreat,],d[indcontr,]) # all matched obs

# ATT and SE for within-cluster matching (W)

attW<-sum( (mwd$sim.lowapg[mwd$sim.cs==1]
-mwd$sim.lowapg[mwd$sim.cs==0]) *weights/sum(weights) )

relattW<-abs(attW-trueatt)/trueatt

m0<-lm(formula = sim.lowapg ~ sim.cs,
data=mwd, weights=c(weights,weights))
m0.vcovCL<-cluster.vcov(m0, mwd$Hospital)
seW<-coefest(m0, m0.vcovCL)[4]

# Average number of controls replicates

ff<-(aggregate(weights ~ indcontr , data = cbind(weights,indcontr) , sum))

arB<-sum(ff[, "weights"][which(ff[, "weights"]>1)]-1)/
length(unique(ff[, "indcontr"]))#in general

# Balance after within-cluster matching

bal_amw = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 +Z
, data=mwd,ks = FALSE,
weights=c(weights,weights), nboots=0, print.level=0)

asamW <-vector()
for(i in 1:18){asamW[[i]] <- bal_amw$BeforeMatching[[i]]$sdiff}
```

```
##### Preferential within-cluster matching (PW)
##### (first matching within-cluster; remaining units matched between clusters)

umwd<-rbind(d[inddrop,],d[which(d$sim.cs==0),])

# Treated not matched within clusters and controls

rrc <- Match(
  Y=umwd$sim.lowapg,
  Tr=umwd$sim.cs,
  X=umwd$sim.ps,
  caliper=caliper*sd(d$sim.ps)/sd(umwd$sim.ps),
  M=1, replace=replacement,ties=TRUE)

mbdata<-rbind(umwd[rrc$index.treated,],umwd[rrc$index.control,])

# Treated matched between (but not within) and matched controls

mwbd<-rbind(mwd,mbdata)

# Treated matched within or between (and matched controls)

# Drops after preferential within match
mwbdndrop<-table(d[rrc$index.dropped,]$ord.Hospital)

# ATT after preferential within-cluster matching
attPW<-sum((mwbd$sim.lowapg[mwbd$sim.cs==1]
-mwbd$sim.lowapg[mwbd$sim.cs==0])
*c(weights,rrc$weights)
/sum(c(weights,rrc$weights)))

#mean(mwbd$sim.lowapg[mwbd$sim.cs==1]
-mwbd$sim.lowapg[mwbd$sim.cs==0])# this only if TIES=FALSE

relattWb<-abs(attPW-trueatt)/trueatt

m0<-lm(formula = sim.lowapg ~ sim.cs,
  data=mwbd,weights=rep(c(weights,rrc$weights),2))
m0.vcovCL<-cluster.vcov(m0, mwbd$Hospital)
sePW<-coefest(m0, m0.vcovCL)[4]

# Average number of controls replicates
indcontrc<-c(indcontr,rrc$index.control)
weightsc<-c(weights,rrc$weights)

ff<-(aggregate(weightsc ~ indcontrc , data = cbind(weightsc,indcontrc) , sum))

arC<-sum(ff[, "weightsc"][which(ff[, "weightsc"]>1)]-1)
/length(unique(ff[, "indcontrc"]))#in general

# Balance after preferential within-cluster matching

bal_amwb = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 +Z,
```

```
data=mwbd,ks = FALSE,
weights=c(weights,weights,rrc$weights,rrc$weights),
nboots=0, print.level=0)

asamPW      <-vector()
for(i in 1:18){asamPW[[i]] <- bal_amwb$BeforeMatching[[i]]$sdiff}

##### Pooled matching with ps from fixed-effects logit model (FE)

# Fixed-effects logit ps

sim.fit2<-glm(formula = sim.cs ~
X1+X2+X3+X4+X5+X6+X7
+X8+X9+X10+X11+ Hospital -1,
data=d, family = binomial(link = "logit"))

d$sim.ps<-fitted(sim.fit2)

# Pooled matching using fixed-effects logit model

rrD = Match(
Y=d$sim.lowapg,
Tr=d$sim.cs,
X=d$sim.ps, caliper=caliper,
M=1, replace=replacement,ties=TRUE)

# Drops
mndropD<-table(d[rrD$index.dropped,]$ord.Hospital)

# Matched data
mdD<-rbind(d[rrD$index.treated,],d[rrD$index.control,])

# ATT and SE
attFE<-rrD$est

relattFE<-abs(attFE-trueatt)/trueatt

# seFE<-rrD$se

m0<-lm(formula = sim.lowapg ~ sim.cs, data=mdD,
weights=c(rrD$weights,rrD$weights)) #summary(m0)
m0.vcovCL<-cluster.vcov(m0, mdD$Hospital)
seFE<-coefest(m0, m0.vcovCL)[4]

# Average number of controls replicates

ff<-(aggregate(rrD$weights ~ rrD$index.control ,
data = cbind(rrD$weights,rrD$index.control) , sum))

arD<-sum(ff[, "rrD$weights"][which(ff[, "rrD$weights"]>1)]-1)/
length(unique(ff[, "rrD$index.control"]))#in general

# Balance after matching using fixed-effects logit model

bal_apmD = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 + Z,
```

```

data=d, match.out=rrD, ks = FALSE,
nboots=0, print.level=0)

asamFE      <-vector()
for(i in 1:18){asamFE[[i]] <- bal_apmD$AfterMatching[[i]]$sdiff}

#### Pooled match with ps from multilevel (random-effects) logit model (RE)

# Random-effects logit model

sim.fit3 <- glmer(formula = sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11+ (1 | Hospital),
data=d, family = binomial(link="logit"))

d$sim.ps<-fitted(sim.fit3)

# Pooled matching with random-effects ps model

rrE = Match(
Y=d$sim.lowapg,
Tr=d$sim.cs,
X=d$sim.ps, caliper=caliper,
M=1, replace=replacement,ties=TRUE)

# Drops
mndropE<-table(d[rrE$index.dropped,]$Hospital)

# Matched dataset
mdE<-rbind(d[rrE$index.treated,],d[rrE$index.control,])

# ATT and SE after matching with method RE
attRE<-rrE$est

relattRE<-abs(attRE-trueatt)/trueatt

# seRE<-rrE$se

m0<-lm(formula = sim.lowapg ~ sim.cs,
data=mdE, weights=c(rrE$weights,rrE$weights)) #summary(m0)
m0.vcovCL<-cluster.vcov(m0, mdE$Hospital)
seRE<-coefest(m0, m0.vcovCL)[4]

# Average number of replicates

ff<-(aggregate(rrE$weights ~ rrE$index.control ,
data = cbind(rrE$weights,rrE$index.control) , sum))

arE<-sum(ff[, "rrE$weights"][which(ff[, "rrE$weights"]>1)]-1)/
length(unique(ff[, "rrE$index.control"]))#in general

# Balance after matching using fixed-effects logit model

bal_apmE = MatchBalance(sim.cs ~
X1+X2+X3+X4+X5+X6+X7+X8+X9+X10+X11 + Z,
data=d, match.out=rrE, ks = FALSE, nboots=0, print.level=0)

```

```
asamRE      <-vector()
for(i in 1:18){asamRE[[i]] <- bal_apmE$AfterMatching[[i]]$sdiff}

##### Udpate iteration results

covnames<-c("X1", "X2", "X3", "X4", "X5",
            "X6", "X7", "X8", "X9", "X10",
            "X11", "X12", "X13",
            "X14", "Z")

strnames<-c("crude", "NV", "W", "PW", "FE", "RE")

drop[[r]]<-t(matrix(
  cbind(mndrop,mwndrop,mwbndrop,mndropD,mndropE),
  nrow=5,ncol=length(mndrop),byrow=TRUE,
  dimnames<-list(c("NV", "W", "PW", "FE", "RE"),
  names(table(d$ord.Hospital))))))

pdrop[[r]]<-apply(sim.drop[[r]],2,sum)/sum(d$sim.cs)

sim.asamsr<-t(matrix(cbind(asam_bef,asamNV,asamW,
asamPW,asamFE,asamRE),nrow=6,ncol=length(asam_bef),
byrow=TRUE,dimnames<-strnames,covnames)))

sim.asams[[r]]<-apply(abs(sim.asamsr),2,mean)
sim.asams.noZ[[r]]<-apply(abs(sim.asamsr[-18,]),2,mean)
sim.asams.Z[[r]]<-abs(sim.asamsr["Z",])

sim.att[[r]]<-c(attbef,attNV,attW,attPW,attFE,attRE,trueatt)
names(sim.att[[r]])<-c(strnames,"true")

sim.relatt[[r]]<-c(relattbef,relattNV,relattW,
relattWb,relattFE,relattRE,trueatt)
names(sim.att[[r]])<-c(strnames,"true")

sim.se[[r]]<-c(sebef,seNV,seW,sePW,seFE,seRE)
names(sim.se[[r]])<-strnames

sim.ar[[r]]<-c(arA,arB,arC,arD,arE)
names(sim.ar[[r]])<-strnames[-1]

}#end cycle on r

##### Create tables with all results

alldrops<-drop[[1]]
for (r in 2:R){alldrops<-rbind(alldrops,drop[[r]])}

allpdrops<-pdrop[[1]]
for (r in 2:R){allpdrops<-rbind(allpdrops,pdrop[[r]])}

allasams<-data.frame(t(matrix(unlist(sim.asams),nrow=6,ncol=R)))
colnames(allasams)<-strnames
```

```

allasamsX<-data.frame(t(matrix(unlist(sim.asams.noZ),nrow=6,ncol=R)))
colnames(allasamsX)<-strnames

allasamsH<-data.frame(t(matrix(unlist(sim.asams.Z),nrow=6,ncol=R)))
colnames(allasamsH)<-strnames

allatts<-data.frame(t(matrix(unlist(sim.att),nrow=7,ncol=R)))
colnames(allatts)<-strnames

allrelatts<-data.frame(t(matrix(unlist(sim.relatt),nrow=7,ncol=R)))
colnames(allrelatts)<-c(stranames,"true")

allse<-data.frame(t(matrix(unlist(sim.se),nrow=6,ncol=R)))
colnames(allse)<-strnames

allar<-data.frame(t(matrix(unlist(sim.ar),nrow=5,ncol=R)))
colnames(allar)<-strnames[-1]

allsquares<-(allatts-allatts[, "true"])^2
colnames(allsquares)<-strnames,"true")

save.image(file=paste("R",R,"betaZ",betaZ,"calip",caliper,
"replacement",replacement,".Rdata",sep=""))

)#end cycle on cht

```

C - Tables with full simulation results

METRICS		STRATEGY [†]					
		Crude	NV	W	PW	FE	RE
$\beta_Z = 0$	No. unmatched units	-	0.52	51.35	0.52	0.35	0.30
	Av. n. of controls replications	-	0.09	0.14	0.14	0.09	0.09
$ICC_Y = 7.1\%$	ASB Z	4.97	1.18	0.00	0.29	0.93	0.91
$ICC_T = 1.1\%$	ASB X	13.22	1.08	1.68	1.67	0.93	0.94
	ASB All	12.76	1.08	1.58	1.59	0.93	0.94
	% Bias	54.15	0.92	2.96	0.17	9.93	9.45
	$RMSE_{montecarlo}$	6.01	3.48	3.43	3.56	3.40	3.29
	SE	2.14	3.56	3.40	3.44	3.40	3.38
$\beta_Z = 0.2$	No. unmatched units	-	0.77	48.27	0.77	0.71	0.70
	Av. n. of controls replications	-	0.10	0.16	0.16	0.10	0.10
$ICC_Y = 7.5\%$	ASB Z	17.61	1.26	0.00	0.45	0.92	0.89
$ICC_T = 2.2\%$	ASB X	13.03	1.13	1.69	1.69	0.95	0.97
	ASB All	13.29	1.13	1.59	1.62	0.95	0.97
	% Bias	59.24	0.11	1.64	1.53	10.05	9.51
	$RMSE_{montecarlo}$	5.36	1.84	1.06	1.48	3.40	2.50
	SE	2.15	3.48	2.95	3.46	3.40	3.28
$\beta_Z = 0.4$	No. unmatched units	-	1.22	40.46	1.22	0.96	1.05
	Av. n. of controls replications	-	0.13	0.19	0.19	0.12	0.12
$ICC_Y = 7.2\%$	ASB Z	35.36	1.23	0.00	0.50	0.86	0.85
$ICC_T = 4.9\%$	ASB X	12.87	1.20	1.76	1.79	1.01	1.02
	ASB All	14.12	1.20	1.66	1.71	1.00	1.01
	% Bias	63.74	0.15	1.39	0.85	9.64	9.36
	$RMSE_{montecarlo}$	7.20	3.76	3.81	3.93	3.70	3.68
	SE	2.15	3.81	3.68	3.48	3.60	3.56
$\beta_Z = 0.6$	No. unmatched units	-	0.94	32.42	0.94	0.66	0.74
	Av. n. of controls replications	-	0.16	0.23	0.23	0.16	0.16
$ICC_Y = 6.7\%$	ASB Z	52.55	1.14	0.00	0.47	0.78	0.80
$ICC_T = 9.0\%$	ASB X	12.72	1.27	1.82	1.87	1.10	1.10
	ASB All	14.93	1.26	1.72	1.79	1.09	1.09
	% Bias	67.70	1.16	1.48	4.24	9.89	10.97
	$RMSE_{montecarlo}$	7.69	4.01	3.96	4.10	3.90	3.90
	SE	2.15	3.82	3.81	3.41	5.71	3.68

[†] NV: naive; W: within-cluster; PW: preferential within-cluster; FE: fixed-effect; RE: random-effect. The proportion of matched pairs that are within clusters in PW is $\approx 99\%$.

Table C1. Simulation results after propensity score matching with replacement when the cluster-level confounder Z is observed. $ATT \approx 10.5\%$.

METRICS		STRATEGY [†]					
		Crude	NV	W	PW	FE	RE
$\beta_Z = 0$ $ICC(Y) = 7.1\%$ $ICC(T) = 11.6\%$	No. unmatched units	-	0.47	49.97	0.47	0.15	0.14
	Av. n. of controls replications	-	0.02	0.07	0.07	0.04	0.04
	ASB Z	30.40	31.58	0.00	0.43	1.02	0.97
	ASB X	13.80	1.31	1.84	1.83	1.12	1.09
	ASB All	14.72	2.99	1.74	1.75	1.11	1.08
	% Bias	69.61	16.93	2.93	3.44	11.63	11.12
	$RMSE_{montecarlo}$	8.70	4.43	4.23	4.45	4.20	4.25
	SE	2.32	4.50	4.15	4.20	4.19	4.19
$\beta_Z = 0.2$ $ICC(Y) = 7.9\%$ $ICC(T) = 13.7\%$	No. unmatched units	-	0.40	53.59	0.40	0.13	0.13
	Av. n. of controls replications	-	0.03	0.09	0.09	0.05	0.05
	ASB Z	47.24	48.39	0.00	0.70	1.00	0.99
	ASB X	13.35	1.31	1.90	1.88	1.15	1.15
	ASB All	15.24	3.92	1.80	1.82	1.14	1.14
	% Bias	72.47	23.86	3.47	4.74	11.94	12.11
	$RMSE_{montecarlo}$	9.17	4.79	4.39	4.57	4.44	4.41
	SE	2.30	4.64	4.18	4.10	4.21	4.24
$\beta_Z = 0.4$ $ICC(Y) = 7.8\%$ $ICC(T) = 17.1\%$	No. unmatched units	-	0.47	57.83	0.47	0.13	0.13
	Av. n. of controls replications	-	0.03	0.11	0.11	0.07	0.07
	ASB Z	63.42	64.37	0.00	1.03	0.97	1.01
	ASB X	12.82	1.28	2.01	1.99	1.23	1.21
	ASB All	15.63	4.79	1.90	1.93	1.21	1.20
	% Bias	75.13	31.47	2.69	6.43	12.64	12.86
	$RMSE_{montecarlo}$	9.54	5.32	4.22	4.41	4.62	4.62
	SE	2.28	4.78	4.20	4.10	4.35	4.35
$\beta_Z = 0.6$ $ICC(Y) = 8.2\%$ $ICC(T) = 21.2\%$	No. unmatched units	-	0.36	63.88	0.36	0.10	0.10
	Av. n. of controls replications	-	0.03	0.15	0.15	0.10	0.10
	ASB Z	78.81	79.56	0.00	1.45	0.95	0.94
	ASB X	12.27	1.24	2.19	2.15	1.33	1.32
	ASB All	15.97	5.59	2.07	2.11	1.31	1.29
	% Bias	77.16	35.67	0.12	9.88	12.46	12.31
	$RMSE_{montecarlo}$	9.80	5.70	4.39	4.76	4.89	4.90
	SE	2.26	4.83	4.17	3.96	4.50	4.48

[†] NV: naive; W: within-cluster; PW: preferential within-cluster; FE: fixed-effect; RE: random-effect. The proportion of matched pairs that are within clusters in PW is $\approx 99\%$.

Table C2. Simulation results after propensity score matching with replacement when coefficients of confounders are increased by 25% in a random subset of clusters and Z is unobserved. $ATT \approx 12\%$.

METRICS		STRATEGY [†]					
		Crude	NV	W	PW	FE	RE
$\beta_Z = 0$	No. unmatched units	-	0.59	442.42	0.59	0.89	0.01
	Av. n. of controls replications	-	0.07	0.28	0.29	0.10	0.08
$ICC_Y = 1.5\%$	ASB Z	1.89	2.05	0.00	0.76	0.91	1.92
$ICC_T = 1.3\%$	ASB X	13.07	0.89	2.60	2.39	0.84	0.84
	ASB All	12.45	0.96	2.46	2.30	0.85	0.90
	%Bias	53.69	0.33	19.52	0.59	8.79	8.82
	$RMSE_{montecarlo}$	5.68	3.24	3.37	3.44	3.21	3.17
	SE	2.09	3.24	2.62	4.38	3.12	3.00
$\beta_Z = 0.2$	No. unmatched units	-	0.64	450.54	0.64	0.87	0.00
	Av. n. of controls replications	-	0.07	0.29	0.30	0.10	0.08
$ICC_Y = 1.5\%$	ASB Z	19.25	19.87	0.00	2.09	0.93	18.29
$ICC_T = 2.4\%$	ASB X	12.96	0.88	2.62	2.39	0.88	0.90
	ASB All	13.31	1.93	2.47	2.38	0.88	1.86
	%Bias	59.81	11.36	19.35	5.34	9.43	18.13
	$RMSE_{montecarlo}$	6.38	3.42	3.36	3.52	3.24	3.56
	SE	2.12	3.24	2.70	4.55	3.29	3.12
$\beta_Z = 0.4$	No. unmatched units	-	0.43	470.18	0.43	0.99	0.04
	Av. n. of controls replications	-	0.07	0.32	0.32	0.12	0.09
$ICC_Y = 1.5\%$	ASB Z	37.31	39.01	0.00	4.56	0.91	35.32
$ICC_T = 5.4\%$	ASB X	12.69	0.89	2.57	2.35	0.96	0.94
	ASB All	14.05	3.01	2.43	2.47	0.96	2.85
	%Bias	65.48	20.96	20.36	8.16	10.52	26.64
	$RMSE_{montecarlo}$	7.12	3.86	3.62	3.58	3.53	4.06
	SE	2.11	3.26	2.75	5.11	3.39	3.04
$\beta_Z = 0.6$	No. unmatched units	-	0.37	312.30	0.37	0.51	0.11
	Av. n. of controls replications	-	0.07	0.36	0.35	0.15	0.09
$ICC_Y = 1.6\%$	ASB Z	54.23	56.78	0.00	7.24	0.94	51.09
$ICC_T = 10.1\%$	ASB X	12.29	0.88	2.59	2.37	1.10	0.95
	ASB All	14.62	3.99	2.45	2.64	1.09	3.74
	%Bias	68.98	27.50	21.29	10.12	8.58	33.04
	$RMSE_{montecarlo}$	7.64	4.27	3.97	3.81	3.99	4.64
	SE	2.11	3.25	2.86	6.50	3.68	3.07

[†] NV: naive; W: within-cluster; PW: preferential within-cluster; FE: fixed-effect; RE: random-effect. The proportion of matched pairs that are within clusters in PW is $\approx 92\%$.

Table C3. Simulation results after propensity score matching with replacement when clusters' sizes are ≈ 100 observations on average and Z is unobserved. $ATT \approx 10.4\%$.

METRICS		STRATEGY [†]					
		Crude	NV	W	PW	FE	RE
<i>Size</i> = 50	No. unmatched units	-	0.63	791.97	0.63	1.39	0.03
	Av. n. of controls replications	-	0.07	0.37	0.28	0.13	0.08
	<i>ICC</i> (<i>Y</i>) = 1.3%	36.29	38.13	0.00	7.22	0.91	36.47
	<i>ICC</i> (<i>T</i>) = 5.4%	12.65	0.88	2.37	2.06	0.94	0.90
	ASB All	13.97	2.95	2.24	2.35	0.94	2.88
	% Bias	62.84	20.57	25.88	9.16	9.09	28.38
	<i>RMSE</i> _{montecarlo}	6.81	3.82	3.89	3.70	3.59	4.10
	<i>SE</i>	2.12	3.14	2.72	1.15	3.30	2.87
<i>Size</i> = 100	No. unmatched units	-	0.43	470.18	0.43	0.99	0.04
	Av. n. of controls replications	-	0.07	0.32	0.32	0.12	0.09
	<i>ICC</i> (<i>Y</i>) = 1.5%	37.31	39.01	0.00	4.56	0.91	35.32
	<i>ICC</i> (<i>T</i>) = 5.4%	12.69	0.89	2.57	2.35	0.96	0.94
	ASB All	14.05	3.01	2.43	2.47	0.96	2.85
	%Bias	65.48	20.96	20.36	8.16	10.52	26.64
	<i>RMSE</i> _{montecarlo}	7.12	3.86	3.62	3.58	3.53	4.06
	<i>SE</i>	2.11	3.26	2.75	5.11	3.39	3.04
<i>Size</i> = 200	No. unmatched units	-	0.82	269.19	0.82	0.84	0.05
	Av. n. of controls replications	-	0.08	0.28	0.30	0.12	0.09
	<i>ICC</i> (<i>Y</i>) = 1.8%	36.10	37.15	0.00	2.46	0.94	32.64
	<i>ICC</i> (<i>T</i>) = 5.2%	13.17	0.90	2.66	2.53	0.96	0.92
	ASB All	14.44	2.91	2.51	2.53	0.96	2.69
	% Bias	61.82	17.02	14.42	5.43	8.44	22.12
	<i>RMSE</i> _{montecarlo}	6.76	3.62	3.36	3.53	3.49	3.85
	<i>SE</i>	2.11	3.16	2.88	3.32	3.27	3.01
<i>Size</i> = 300	No. unmatched units	-	0.42	191.97	0.42	0.63	1.08
	Av. n. of controls replications	-	0.08	0.24	0.25	0.12	0.10
	<i>ICC</i> (<i>Y</i>) = 2.9%	33.87	34.64	0.00	1.37	0.98	26.26
	<i>ICC</i> (<i>Y</i>) = 5.1%	12.99	0.94	2.50	2.38	1.00	0.96
	ASB All	14.44	2.91	2.51	2.53	0.96	2.69
	% Bias	63.15	19.01	12.03	4.59	10.92	21.83
	<i>RMSE</i> _{montecarlo}	6.76	3.62	3.36	3.53	3.49	3.85
	<i>SE</i>	2.11	3.16	2.88	3.32	3.27	3.01

[†] NV: naive; W: within-cluster; PW: preferential within-cluster; FE: fixed-effect; RE: random-effect.

Table C4. Simulation results after propensity score matching with different sizes of clusters. *Z* is unobserved. *ATT* $\approx$ 10.4%

METRICS		STRATEGY [†]					
		Crude	NV	W	PW	FE	RE
$ICC(X') \approx 10\%$	No. unmatched units	-	0.12	26.31	0.12	0.08	0.03
	Av. n. of controls replications	-	0.08	0.14	0.14	0.09	0.09
	ASB Z	41.99	2.02	0.00	0.23	0.79	0.74
	ASB X	13.27	2.71	1.90	1.92	1.00	1.00
	ASB All	14.86	2.67	1.80	1.83	0.99	0.99
	% Bias	76.87	17.26	13.55	14.11	6.94	6.77
	$RMSE_{montecarlo}$	11.33	4.62	4.50	4.61	4.09	4.11
	SE	2.46	4.64	4.67	5.30	4.62	4.63
$ICC(X') \approx 30\%$	No. unmatched units	-	0.47	32.66	0.47	1.07	1.14
	Av. n. of controls replications	-	0.12	0.18	0.18	0.14	0.14
	ASB Z	41.67	10.24	0.00	0.27	0.81	0.90
	ASB X	12.69	2.90	2.21	2.27	1.05	1.05
	ASB All	14.30	3.31	2.09	2.16	1.04	1.05
	% Bias	65.23	25.69	18.29	20.04	8.73	8.38
	$RMSE_{montecarlo}$	8.12	4.71	4.23	4.38	4.12	3.90
	SE	2.28	3.53	3.52	3.66	3.72	3.66

[†] NV: naive; W: within-cluster; PW: preferential within-cluster; FE: fixed-effect; RE: random-effect. $ICC(X') \approx \{10\%, 30\%\}$ were obtained by setting $X' = X + 0.6Z + \eta_j$ and $\sigma_{\eta} = \{0.2, 0.6\}$. X is the variable gestational age. $ICC(Y)$ and $ICC(T)$ are as in Table C1 for $\beta_Z = 0.4$.

Table C5. Simulation results after propensity score matching with a positive $ICC(X')$. $ATT \approx 14.7\%$